PhysiXDeform: Real-Time Vision-Guided Soft-Tissue Deformation Prediction With Physical Priors

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***Abstract*—Accurate prediction of soft tissue deformation from endoscopic video is critical for robot-assisted, image-guided interventions. However, it remains challenging due to occlusions, complex dynamics, and the absence of direct physical measurements. Existing vision-based approaches often impose rigid or simplified motion priors, limiting their generalisability and physical interpretability. In this letter, we introduce a vision-guided, physics-informed framework that predicts multi-step 3D deformation by integrating stereo disparity, optical flow, and tracked keypoints into a spatio-temporal graph. Motion features, including velocity, acceleration, and directional strain, are encoded as node and edge attributes and processed via a graph network. The model is trained end-to-end with self-supervised objectives promoting geometric alignment, temporal coherence, and motion consistency. Across synthetic and real surgical datasets, our method achieves up to 28.8% reduction in Chamfer Distance (CD) at short prediction horizons, and an average 43.9% reduction over extended-horizon forecasts. In cross-scene adaptation, our model achieves average CD reductions of 12.9% with light tuning and 12.2% with heavy tuning, where the latter shows slight overfitting but still outperforms mesh-based baselines and supports real-time operation.**

***Index Terms*—Computer vision for medical robotics, computer vision for automation, medical robots and systems.**

I. Introduction

ENDOSCOPY is central to modern clinical practice and robot-assisted interventions, enabling direct visualisation of internal anatomy for diagnostic and therapeutic purposes. However, minimally invasive and robot-assisted procedures must contend with continuous, non-rigid soft-tissue deformation caused by respiration, peristalsis, and patient motion. Combined with occlusions, low-texture surfaces, and visual ambiguity, these dynamics make tissue motion difficult to model and often lead to errors in localisation, depth estimation, and 3D reconstruction, resulting in inaccurate tool trajectories. Real-time, vision-based deformation modelling is therefore essential for robust surgical robotics, enabling deformation-aware perception and image-guided navigation under deforming anatomy.

Most existing approaches in surgical and endoscopic perception rely on rigid or simplified deformation assumptions, approximating tissue motion with static, affine, or piecewise-rigid models [1]. Such formulations fail to capture the non-linear, time-dependent behaviour of soft tissues in vivo under dynamic and visually ambiguous conditions, leading to geometric inconsistencies, localisation drift, and reconstruction artefacts over extended horizons. Although some recent work introduces deformation-aware components [2], many methods remain constrained by rigid-world assumptions.

Physics-based methods, such as finite element models (FEM), simulate non-linear, viscoelastic soft-tissue deformation with high biomechanical fidelity [3]. However, they require detailed meshes, calibrated material parameters, and computationally expensive solvers, limiting real-time deployment on typical robotic hardware. Related acceleration strategies remain largely restricted to offline or task-specific settings and therefore fall short of the low-latency demands of robot-assisted surgery.

In this letter, we propose a vision-guided, physics-informed deformation prediction model for minimally invasive surgery and capsule endoscopy that relies solely on on-board cameras. The proposed approach models spatially consistent and temporally coherent soft-tissue motion using a graph network [4], generalising across anatomical regions while preserving physical plausibility. By embedding physics-inspired motion priors directly into the learning pipeline, the model enables robust, multi-step deformation prediction under occlusion, noise, and visual ambiguity, suitable for real-time deployment.

Our main contributions are as follows:

- We introduce the first *vision-guided* deformation prediction framework for surgical robotics and endoscopic applications in camera-only clinical settings.
- We design physics-informed regularisation comprising acceleration-consistency prior and patch-flow alignment to promote biomechanical plausibility and improve generalisation across anatomically diverse surgical scenes.
- We demonstrate robust generalisation across synthetic and clinical datasets, with empirical validation on gastrointestinal, cardiac, hepatic, and abdominal procedures, achieving

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sub-millimetre accuracy and real-time performance under diverse conditions.

II. Related Work

A. Classical Deformation Models

Classical deformation models are widely used in surgical simulation and vision due to their computational efficiency and geometric simplicity. Embedded deformation and ARAP approximate tissue motion using locally rigid or affine transforms, preserving structure but limiting non-linear flexibility [5]. Mass-spring, shape matching, and position-based dynamics enable real-time updates but are often parameter-sensitive and lack biomechanical fidelity under large deformations [6], [7]. Non-rigid structure-from-motion (NRSfM) estimates deformation from monocular views under low-rank or isometric priors, but degrades in the presence of rapid or complex motion [8].

B. Learning-Based Deformation Prediction

Graph networks have been widely used as data-driven physical models for deformation prediction. GNS and MeshGraphNet learn mesh-based interactions via message passing to simulate fluids, cloth, and elastic solids [9], [10]. In surgical settings, learning-based predictors trained on FEM data have been explored but typically rely on known geometry and offline meshing, limiting clinical applicability [11]. Related work in robotics largely focuses on cloth and deformable linear objects, including RGB-based forecasting with differentiable 3D representations and diffusion-based partial-state estimation [12], [13]. These methods address externally actuated, structured manipulation, whereas we target passive soft-tissue deformation from vision without ground truth.

C. Soft-Tissue Deformation Modelling

Several approaches model soft-tissue deformation in endoscopy with an emphasis on visual realism and view consistency. Prior work reconstructs deformable surfaces using dynamic implicit fields or point-based representations from stereo or RGB-D video [14], [15]. These methods prioritise visually coherent rendering but provide weakly constrained, non-metric geometry. In contrast, we pursue physically informed, image-driven deformation prediction with metric accuracy for clinical applications.

D. Physics-Based Simulation for Surgical Deformation

Physics-based simulation is widely used for modelling soft-tissue deformation in surgical settings. FEM enable accurate continuum mechanics modelling and have been applied to surgical simulation and planning under nonlinear deformation [16]. To improve efficiency, Position-Based Dynamics (PBD) provides stable real-time simulation and has been used for neurosurgical deformation modelling and intra-operative control [17], [18]. Recent work combines physical simulation with vision through real-to-sim registration or parameter optimisation to align simulated tissue with observed motion for robotic autonomy [19], [20]. Simulation frameworks have also been proposed for modelling tool–tissue interaction and cutting dynamics [21]. While physically interpretable, these approaches rely on explicit simulators and material parameters, limiting their applicability to vision-only, real-time deformation prediction.

III. Methodology

The proposed framework predicts multi-step soft-tissue deformation from stereo endoscopic sequences (Fig. 1). Given three consecutive stereo pairs $\{\mathbf{I}_{t-2}, \mathbf{I}_{t-1}, \mathbf{I}_t\}$, grid-sampled 2D keypoints are tracked across frames and reprojected using stereo disparity to form a sparse 3D mesh $\mathbf{P}_t$, while optical flow provides complementary motion cues. Each node is encoded with position, velocity, acceleration, and local flow, and each edge with directional strain. These features are processed by a physics-informed graph network with an encoder–processor–decoder architecture and seven message-passing steps to predict the next mesh state $\hat{\mathbf{P}}_{t+1}$. The model is trained in a self-supervised manner using position, acceleration, and patch-flow consistency losses from $\mathbf{I}_{t+1}$.

A. Visual and Motion Feature Extraction

A regular 2D grid (e.g., 10×10) is applied to the reference frame $\mathbf{I}_t$ to sample candidate keypoints, excluding a boundary margin to avoid artefacts. Disparity maps are computed for each stereo pair [22], and sampled keypoints are reprojected into 3D using the corresponding stereo geometry. Keypoints with extreme depth values are removed to mitigate errors from shallow viewing angles or distant background regions. The remaining keypoints are tracked across frames [23], and those affected by occlusion, boundary violations, or incomplete trajectories are discarded to maintain spatio-temporal consistency. In parallel, dense optical flow is estimated between adjacent frames [24]; for each keypoint, we extract the optical-flow vector $\mathbf{f}_t^i \in \mathbb{R}^2$ and retain a centred $p \times p$ flow patch *only* for the alignment regulariser in Section III-C. At each time step t, the tracked keypoints are organised into a sparse mesh by applying constrained Delaunay triangulation (CDT) in the image plane, enforcing non-intersecting edges to preserve spatial locality. Since each mesh node corresponds to a tracked keypoint with an associated 3D position from stereo reconstruction, the resulting 2D connectivity is directly lifted to 3D, yielding a sparse 3D mesh. The 3D node positions are denoted by $\mathbf{P}_t = [\,\mathbf{p}_t^1, \ldots, \mathbf{p}_t^N\,]^\top$, where N is the number of nodes. At each node, motion features are computed, including velocity $\mathbf{v}_t^i = \mathbf{p}_t^i - \mathbf{p}_{t-1}^i$, acceleration $\mathbf{a}_t^i = \mathbf{p}_t^i - 2\mathbf{p}_{t-1}^i + \mathbf{p}_{t-2}^i$, and the optical-flow vector $\mathbf{f}_t^i$, thereby defining graph nodes with motion attributes derived from visual observations.

B. Physics-Informed Graph Prediction Module

1) Physics-Informed Graph Construction: Building on these acquired graph attributes, we organise them into physically meaningful node and edge representations.

a) Node Features: Each graph node corresponds to a tracked 3D point and is assigned a spatio-temporal feature vector defined as $\mathbf{x}_t^i = [\,\mathbf{p}_t^i,\ \mathbf{v}_t^i,\ \mathbf{a}_t^i,\ \mathbf{f}_t^i\,]$, where $\mathbf{p}_t^i$ denotes the 3D position, $\mathbf{v}_t^i$ and $\mathbf{a}_t^i$ are the per-node velocity and acceleration, and $\mathbf{f}_t^i$ is a

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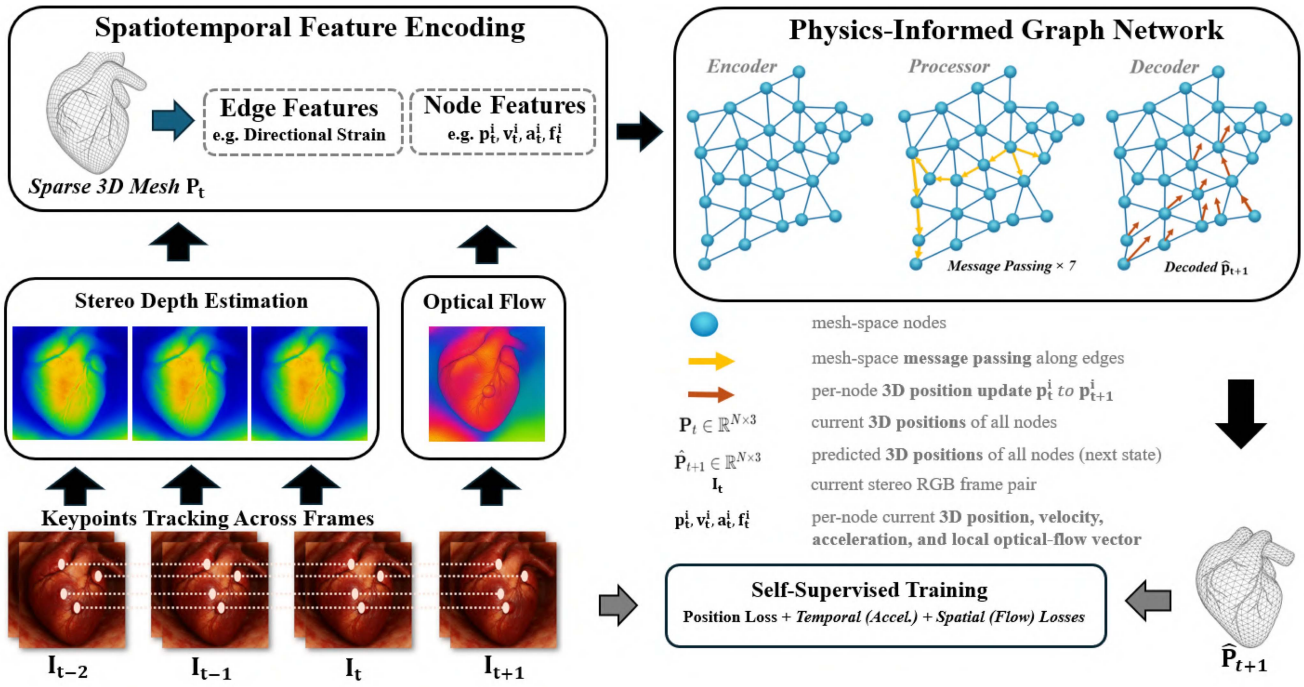


Fig. 1. Overview of the vision-guided, physics-informed predictor. From a sequence of stereo frame pairs $\{\mathbf{I}_{t-2}, \mathbf{I}_{t-1}, \mathbf{I}_t\}$, we track grid-sampled 2D keypoints and compute stereo depth and optical flow, forming a sparse 3D mesh $\mathbf{P}_t$ with per-node features $[\mathbf{p}_t^i, \mathbf{v}_t^i, \mathbf{a}_t^i, \mathbf{f}_t^i]$ and directional strain. A physics-informed graph network (encoder–processor–decoder; 7 message-passing steps) outputs per-node updates $\mathbf{p}_t^i \rightarrow \hat{\mathbf{p}}_{t+1}^i$, yielding $\hat{\mathbf{P}}_{t+1}$. Training is self-supervised with position, temporal (acceleration), and spatial (flow) losses computed from $\mathbf{I}_{t+1}$.

local optical-flow vector. Incorporating both flow and velocity introduces a cross-modal constraint between 2D image-space motion and 3D displacement, encouraging consistent motion cues across domains. Meanwhile, acceleration encodes local temporal dynamics and provides a prior that aids regularisation during training.

b) Edge Features: Edge features encode spatial, temporal, and cross-domain relationships between neighbouring nodes defined by the mesh connectivity. For each edge (i, j), we compute relative displacements in world space $\Delta\mathbf{p}_t^{ij,\text{world}} = \mathbf{p}_t^{i,\text{world}} - \mathbf{p}_t^{j,\text{world}}$, mesh space $\Delta\mathbf{p}_t^{ij,\text{mesh}}$, and image space $\Delta\mathbf{u}_t^{ij} = (\Delta u_t^{ij}, \Delta v_t^{ij})$. Here, world space denotes 3D positions; mesh space is a fixed intrinsic coordinate system that remains constant over time, providing a stable structural prior; and image space corresponds to 2D pixel coordinates. Together, these terms establish aligned 3D–2D geometric relationships. Additionally, relative motion features between neighbouring nodes, including velocity $\Delta\mathbf{v}_t^{ij}$, acceleration $\Delta\mathbf{a}_t^{ij}$, and flow difference $\Delta\mathbf{f}_t^{ij}$, capture local consistency across spatial and temporal domains.

To encode physically meaningful strain-like signals, we normalise world-relative displacement to unit edge direction:

$$\hat{\mathbf{r}}_t^{ij} = \frac{\Delta\mathbf{p}_t^{ij,\text{world}}}{\|\Delta\mathbf{p}_t^{ij,\text{world}}\|} \tag{1}$$

The velocity and acceleration differences are also projected onto this axis:

$$v_t^{ij,\|} = \langle \Delta\mathbf{v}_t^{ij}, \hat{\mathbf{r}}_t^{ij} \rangle, \qquad a_t^{ij,\|} = \langle \Delta\mathbf{a}_t^{ij}, \hat{\mathbf{r}}_t^{ij} \rangle \tag{2}$$

These projections quantify directional strain: positive values indicate local expansion and negative values indicate compression, providing a strong physical prior within the graph.

2) Graph-Based Deformation Prediction Network

Our graph-based prediction module follows the encoder–processor–decoder architecture of MeshGraphNet [9], adapted for vision-centric input and local soft-tissue deformation in clinical settings. The design emphasises spatial locality, incorporate motion-derived features, and maintain robustness in sparse and partially observed mesh sequences.

a) Encoder: The previously constructed graph, consisting of spatio-temporal node and edge features, is passed through an encoder module that embeds each feature into a shared latent space using learned multilayer perceptrons (MLPs). The resulting latent representations $\mathbf{h}_t^{i,(0)}$ and $\mathbf{e}_t^{ij,(0)}$ serve as initial inputs to the graph processor at each time step, enabling local spatio-temporal reasoning.

b) Processor: The processor performs message passing over the encoded graph to model local interactions between mesh nodes. Unlike simulation-based settings such as [9], which assume globally smooth deformation, our clinical scenarios exhibit local and heterogeneous tissue motion, where neighbouring regions may move asynchronously or remain static due to biomechanical constraints and partial observability. We therefore restrict message passing to spatially local interactions and employ an adaptive number of steps K (set to $K = 7$ in our experiments) to balance geometric context with robustness to noise. Excessive propagation can amplify spurious updates across

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asynchronously moving regions, leading to over-smoothing or inconsistent deformation estimates, whereas limiting the receptive field preserves spatial precision and improves robustness under artefacts and partial observability.

At each round k, the hidden state of each node is updated using MLP-based aggregation from its spatial neighbours, based on the latent node features $\mathbf{h}_t^{i,(k-1)}$ and edge features $\mathbf{e}_t^{ij,(k-1)}$:

$$\mathbf{h}_t^{i,(k)} = \phi_{\text{node}}\left(\mathbf{h}_t^{i,(k-1)}, \times \sum_{j\in\mathcal{N}(i)} \phi_{\text{edge}}\left(\mathbf{h}_t^{i,(k-1)}, \mathbf{h}_t^{j,(k-1)}, \mathbf{e}_t^{ij,(k-1)}\right)\right) \tag{3}$$

where $\mathcal{N}(i)$ denotes the set of mesh neighbours of node i, and $\phi_{\text{node}}(\cdot)$ and $\phi_{\text{edge}}(\cdot)$ are shared MLPs. The final node representations $\mathbf{h}_t^{i,(K)}$ are passed to the decoder to produce deformation predictions.

c) Decoder: The decoder produces a single output per node: the predicted acceleration $\hat{\mathbf{a}}_{t+1}^i$. This output serves two purposes. During training, it acts as a regularising signal to encourage smooth and physically consistent motion. At inference time, the predicted acceleration is integrated via a second-order finite-difference update to compute the future 3D position of each node:

$$\hat{\mathbf{p}}_{t+1}^i = 2\mathbf{p}_t^i - \mathbf{p}_{t-1}^i + (\Delta t)^2 \hat{\mathbf{a}}_{t+1}^i, \tag{4}$$

where $\mathbf{p}_t^i$ and $\mathbf{p}_{t-1}^i$ denote the current and previous node positions, respectively, and Δt denotes the temporal interval between consecutive frames. In this formulation, time is discretised at the video frame rate.

C. Self-Supervised Training Strategy

The model is trained end-to-end using a self-supervised objective that enforces temporal coherence, physical plausibility, and image-space motion consistency. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \frac{1}{N}\sum_{i=1}^{N}\left(\|\hat{\mathbf{p}}_{t+1}^i - \mathbf{p}_{t+1}^i\|_2^2 + \lambda_{\text{acc}}\|\hat{\mathbf{a}}_{t+1}^i - \mathbf{a}_t^i\|_2^2 + \lambda_{\text{flow}}\|\hat{\mathbf{f}}_{t+1}^i - \bar{\mathbf{f}}_{t+1}^i\|_2^2\right) \tag{5}$$

where each term corresponds to a distinct component of the self-supervised training signal, and $\|\cdot\|_2$ denotes the Euclidean (L_2) norm.

a) Position Loss: This term supervises the predicted 3D positions $\hat{\mathbf{p}}_{t+1}^i$ using stereo-derived pseudo-ground-truth positions $\mathbf{p}_{t+1}^i$, obtained by projecting tracked 2D keypoints into 3D with calibrated stereo geometry, encouraging alignment between predicted deformation and observed motion.

b) Acceleration Regularisation: An acceleration-based regularisation term enforces temporal smoothness by penalising sudden changes in second-order motion. It compares the predicted acceleration $\hat{\mathbf{a}}_{t+1}^i$ with the observed acceleration $\mathbf{a}_t^i = \mathbf{p}_t^i - 2\mathbf{p}_{t-1}^i + \mathbf{p}_{t-2}^i$, promoting physically plausible tissue dynamics.

c) Flow Alignment Regularisation: To encourage spatially coherent deformation, the projected 3D displacement $\hat{\mathbf{f}}_{t+1}^i = \Pi(\hat{\mathbf{p}}_{t+1}^i - \mathbf{p}_t^i)$ is aligned with the average dense optical flow $\bar{\mathbf{f}}_{t+1}^i$ over a centred $p \times p$ image patch, where $\Pi(\cdot)$ denotes perspective projection using calibrated camera intrinsics. This regulariser discourages unrealistic local deviations and reinforces spatial consistency.

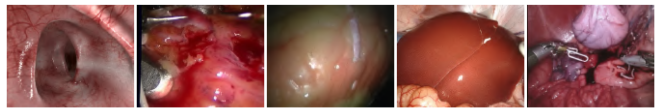


Fig. 2. Representative input frames from each evaluation scenario (left to right): GI (Synthetic), Heart (Hamlyn), Silicone Heart (Hamlyn), Liver (StereoMIS), Abdomen (SurgT).

TABLE I
SUMMARY OF DATASETS FOR TRAINING AND EVALUATION

Dataset	Organ	Modality	FPS	Deformation
GI Sim. [7]	GI tract	Synthetic	1	Sinusoidal
Hamlyn 4 [25]	Heart	MIS (in vivo)	25	Cardiac
Hamlyn 5 [25]	Heart	MIS (in vivo)	25	Cardiac
Hamlyn 11 [26]	Heart	Phantom (Si.)	25	Periodic
StereoMIS P2 [8]	Liver (porcine)	Lap. (in vivo)	30	Respiratory
SurgT 6 [27]	Abd. (porcine)	Lap. (in vivo)	30	Organ shift
SurgT 7 [27]	Abd. (porcine)	Lap. (in vivo)	30	Organ shift

Si. = silicone; Lap. = laparoscopic; MIS = minimally invasive surgery

TABLE II
PARAMETERS IN THE GRAPH-BASED MODEL

Parameter	Value / Description
Message passing steps	7 (shared weights across steps)
MLP depth	2 layers per module
Latent dimension	128
Activation function	ReLU
Flow patch size	6×6 (2 channels)
Node feature size	89
Edge feature size	23

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: We evaluate on synthetic and real-world datasets covering diverse organs, deformation patterns, and imaging conditions. The synthetic GI dataset from [28] simulates anatomically plausible, spatially varying sinusoidal motion and provides stereo image pairs with depth for structured evaluation. For real-world validation, we use three stereo endoscopic datasets: (i) Hamlyn [29], including in vivo cardiac procedures and a periodically actuated silicone heart phantom; (ii) StereoMIS [30], capturing porcine liver deformation under specularities and occlusions; and (iii) SurgT [31], involving porcine abdominal surgery. Sample scenes are shown in Fig. 2, and dataset statistics are summarised in Table I. As no public surgical dataset provides metric in vivo 3D soft-tissue deformation ground truth, we adopt self-supervised training with stereo-derived pseudo-ground truth for evaluation.

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TABLE III
CD (MM) AND PASS RATE (%) AT FIXED THRESHOLDS ACROSS PREDICTION STEPS. MGN = MESHGRAPHNET

Model	Step 1		Step 3		Step 5		Step 7		Step 9	
Synthetic GI										
	CD↓	**< 0.15 mm ↑**	**CD↓**	**< 0.80 mm ↑**	**CD↓**	**< 1.50 mm ↑**	**CD↓**	**< 3.00 mm ↑**	**CD↓**	**< 3.50 mm ↑**
Harmonic	5.643±2.430	0.0	8.683±3.396	0.0	8.146±1.725	0.0	7.594±2.236	0.0	8.631±2.467	0.0
MGN (200k)	**0.106±0.024**	**97.7**	1.594±0.699	7.0	5.102±2.247	0.0	7.819±3.580	9.3	11.682±6.762	2.3
MGN (400k)	0.116±0.028	86.0	1.499±0.558	7.0	5.208±2.316	4.7	8.097±3.256	2.3	10.432±5.095	2.3
Ours (200k)	0.118±0.045	81.4	0.690±0.222	**69.8**	1.076±0.351	88.4	1.717±0.747	95.3	**2.353±0.907**	**86.0**
Ours (400k)	0.111±0.052	81.4	**0.683±0.215**	67.4	**1.027±0.324**	**90.7**	**1.498±0.732**	**95.3**	2.387±1.204	86.0
Hamlyn Heart MIS (Trj.4)										
	CD↓	**< 0.35 mm ↑**	**CD↓**	**< 1.00 mm ↑**	**CD↓**	**< 1.00 mm ↑**	**CD↓**	**< 3.00 mm ↑**	**CD↓**	**< 3.50 mm ↑**
Harmonic	4.222±3.148	0.0	12.49±9.668	0.0	14.06±10.99	0.0	14.52±13.59	0.0	13.37±9.582	0.0
MGN (200k)	0.551±0.278	27.9	1.665±1.522	46.5	1.896±0.610	14.0	4.007±3.318	44.2	5.021±2.602	25.6
MGN (400k)	0.584±0.268	16.4	1.690±1.543	48.8	1.815±1.041	16.3	3.729±2.769	46.5	4.525±2.007	37.2
Ours (200k)	0.338±0.233	76.7	0.896±0.475	62.8	0.998±0.218	51.2	2.759±1.237	55.8	5.216±2.283	25.6
Ours (400k)	**0.326±0.208**	**81.4**	**0.746±0.369**	**79.1**	**0.833±0.383**	**90.7**	**1.871±0.886**	**88.4**	**3.119±1.781**	**76.7**
Hamlyn Silicone Phantom (Trj.11)										
	CD↓	**< 0.30 mm ↑**	**CD↓**	**< 0.35 mm ↑**	**CD↓**	**< 0.50 mm ↑**	**CD↓**	**< 0.70 mm ↑**	**CD↓**	**< 1.50 mm ↑**
Harmonic	2.012±1.882	7.1	5.483±8.772	0.0	7.908±5.202	0.0	5.952±1.603	0.0	18.31±11.56	0.0
MGN (200k)	0.293±0.130	65.1	0.424±0.434	81.4	0.742±0.625	48.8	1.742±1.967	55.8	1.915±1.623	53.5
MGN (400k)	0.301±0.134	51.2	0.415±0.489	83.7	**0.514±0.531**	**76.7**	0.964±1.370	**81.4**	2.033±2.032	58.1
Ours (200k)	0.266±0.101	60.5	0.407±0.200	58.1	0.783±0.601	44.2	1.042±0.863	46.5	1.676±1.503	67.4
Ours (400k)	**0.214±0.084**	**74.4**	**0.250±0.084**	**93.0**	0.615±0.460	48.8	**0.703±0.546**	62.8	**1.123±1.012**	**76.7**
StereoMIS Porcine Liver (Trj.P2)										
	CD↓	**< 0.25 mm ↑**	**CD↓**	**< 0.30 mm ↑**	**CD↓**	**< 0.70 mm ↑**	**CD↓**	**< 1.00 mm ↑**	**CD↓**	**< 1.50 mm ↑**
Harmonic	1.904±2.819	26.2	6.418±9.353	0.0	15.42±22.75	0.0	33.03±49.72	0.0	56.36±82.67	0.0
MGN (200k)	0.263±0.208	58.1	0.401±0.331	55.8	0.867±0.759	62.8	1.047±0.931	74.4	1.397±1.220	79.1
MGN (400k)	0.267±0.236	55.8	0.480±0.415	48.8	0.885±0.581	53.5	1.185±0.597	46.5	1.841±0.877	37.2
Ours (200k)	0.221±0.148	69.8	0.339±0.248	65.1	0.536±0.382	69.8	0.707±0.393	76.7	1.147±0.764	62.8
Ours (400k)	**0.217±0.166**	**72.1**	**0.288±0.198**	**69.8**	**0.492±0.374**	**72.1**	**0.579±0.394**	**86.0**	**0.903±0.624**	**86.0**

2) Training and Evaluation Protocol: Human organ deformation, such as respiration and cardiac motion, typically follows periodic patterns with moderate cycle-to-cycle variation. To capture intra-cycle dynamics without overfitting, we train on the first 500 frames of each dataset, spanning 25–35 motion cycles. Sequences of 17 frames (roughly one cycle) are sampled at varying mesh grid resolutions (10×10 to 19× 19, stride 3) to promote spatial diversity. Evaluation is performed on six 10-frame evaluation sequences starting after frame 500, using a sliding-window strategy with an overlap of 1–5 frames to simulate real-time prediction and mitigate error drift. The temporal duration of each prediction step depends on the dataset frame rate: one step corresponds to 1 s for the Synthetic GI dataset, approximately 40 ms for the Hamlyn datasets, and approximately 33 ms for the StereoMIS and SurgT datasets. To test extended-horizon robustness, we include additional sequences starting at frame 700 or later. Evaluation covers both seen and unseen grid resolutions, with variation introduced by disparity filtering and tracking noise. For all experiments, we fix the regularisation weights to $\lambda_{\text{acc}} = 0.005$ and $\lambda_{\text{flow}} = 0.01$.

We further assess cross-scene generalisation by fine-tuning models pretrained on multiple motion cycles using the same temporally extended training setup. Fine-tuning is performed on separate target sequences from the same organ category, and evaluation is carried out on trajectories that are unseen during both training stages. This setting tests the model's ability to transfer deformation priors across anatomically consistent but visually distinct environments. For fair comparison, we additionally include direct optimisation baselines trained from scratch on the target trajectories only, using the same architecture and training protocol as the fine-tuning stage. Quantitative comparisons between scratch and cross-scene adapted models are reported in Table V. Key architectural settings are summarised in Table II.

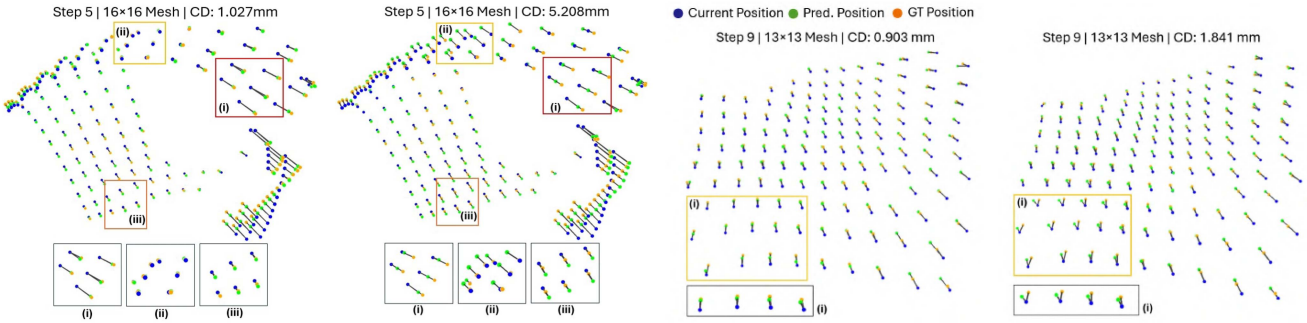
3) Evaluation Metric: We use Chamfer Distance (CD), computed on the sparse set of tracked 3D keypoints (mesh nodes), to evaluate geometric alignment between predicted and reference 3D deformed structures. CD computes the average bidirectional nearest-neighbour distance between two point clouds, capturing global shape similarity while being robust to outliers. This makes it well-suited for our setting, where deformation must be assessed under sparse tracking, partial observability, and visual noise. Compared to pixel-wise or mesh-level metrics, CD offers a practical balance between fidelity and robustness across various conditions.

4) Baseline: To the best of our knowledge, no existing vision-only deformation prediction framework directly aligns with our setting. We therefore compare against MeshGraphNet [9], a strong mesh-based spatio-temporal model and the closest available prior, despite not being designed for vision-guided inference. In addition, we include a harmonic fitting baseline [32] that estimates the dominant frequency using the Fast Fourier Transform (FFT) and fits a compact multi-harmonic trajectory to the 3D displacements.

B. Experimental Results

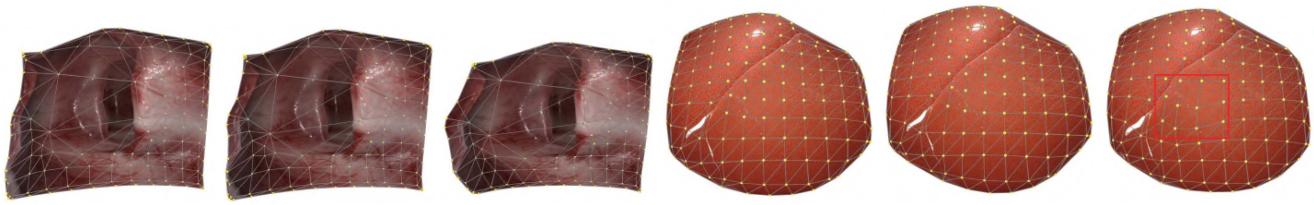
1) Temporally Extended Forecasting: Table III reports CD across four datasets and prediction steps (1–9). At the 400 k checkpoint, our method consistently outperforms MeshGraphNet across all datasets, with particularly strong gains at extended horizons. On the synthetic GI benchmark, CD is reduced by over 50% at Step 3 and over 80% at Step 7, reaching 1.498 mm.

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(a) GI, Ours (b) GI, MeshGraphNet (c) Liver, Ours (d) Liver, MeshGraphNet

Fig. 3. Qualitative comparison of predicted (green) and ground-truth (orange) vertex positions, with blue points denoting current positions. Zoom-in boxes highlight divergence and motion reversal effects.



(a) GI, GT (b) GI, Ours (c) GI, MGN (d) Liver, GT (e) Liver, Ours (f) Liver, MGN

Fig. 4. Textured mesh renderings at Step 20 on GI and porcine liver, comparing ground truth (GT), our model, and MeshGraphNet (MGN). Red boxes highlight local distortions in the predicted meshes.

Averaged across all datasets, our approach achieves a 47.5% relative CD reduction at Step 3, increasing to 52.4% at Step 7. In real-world scenes, substantial improvements are observed at longer horizons, including reductions of 27.1% on Hamlyn Silicone Phantom and 51.1% on Porcine Liver at Step 7, with consistent gains maintained up to Step 9. These results demonstrate robust extended-horizon deformation prediction across diverse settings.

Fig. 5 illustrates CD progression across prediction steps on the GI and Porcine Liver datasets. In both cases, our method shows a consistently slower growth in error compared to MeshGraphNet. On GI, the baseline CD rises steeply, while our curve remains comparatively flat. A similar trend appears in the Porcine Liver dataset, where the baseline increases sharply during early steps, whereas our method maintains lower CD with more stable growth.

To illustrate short- and extended-horizon performance, Figs. 3 and 4 present point-based and textured mesh comparisons on synthetic GI and porcine liver sequences. In the point-based visualisation (Fig. 3), our method produces coherent predictions under large deformation, whereas MeshGraphNet exhibits reversed motion, scattered points, and boundary errors in the highlighted regions. In the rendered results (Fig. 4), our predictions closely match the ground truth in both geometry and texture, while MeshGraphNet introduces noticeable distortion, including stretching in GI and local mesh irregularities in the liver case, leading to warped geometry and texture discontinuities. These qualitative observations are consistent with the

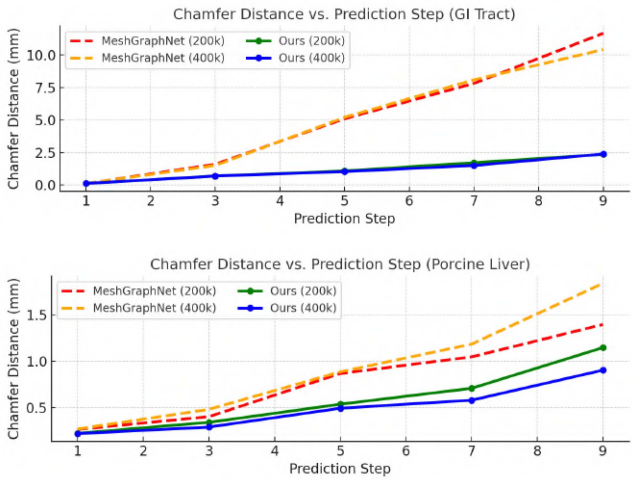


Fig. 5. CD across prediction steps on GI (above) and Porcine Liver (below).

TABLE IV
EXTENDED-STEP CD (MM) ACROSS GI AND LIVER DATASETS

Model	Synthetic GI		StereoMIS Porcine Liver	
	Step 17	Step 20	Step 17	Step 20
	CD↓	CD↓	CD↓	CD↓
MGN (200k)	19.162±5.034	21.957±6.120	5.382±1.808	6.405±1.999
MGN (400k)	18.538±7.574	20.867±10.249	5.049±1.138	6.054±1.550
Ours (200k)	3.813±0.604	4.976±0.864	3.832±1.752	4.718±2.156
Ours (400k)	**2.904±0.970**	**4.006±1.785**	**2.052±0.979**	**2.450±1.184**

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TABLE V
CROSS-SCENE ADAPTATION. CD (MM) AND PASS RATE (%) AT FIXED THRESHOLDS. MGN = MESHGRAPHNET

Model	Step 1		Step 3		Step 7		Step 9	
	CD↓	< 0.15 mm ↑	CD↓	< 0.50 mm ↑	CD↓	< 1.50 mm ↑	CD↓	< 3.00 mm ↑
Hamlyn Heart								
MGN (300k+150k)	0.137±0.067	54.8	0.518±0.361	57.1	1.611±1.307	59.5	2.636±2.115	69.0
MGN (400k+300k)	0.148±0.069	50.0	0.580±0.375	57.1	1.279±0.781	76.2	1.935±1.337	83.3
Ours (Scratch, 150k)	0.110±0.057	71.4	0.369±0.183	71.4	3.217±2.213	35.7	5.571±3.558	35.7
Ours (Scratch, 300k)	0.116±0.061	69.0	0.364±0.187	76.2	3.766±2.673	33.3	6.380±4.155	35.7
Ours (300k+150k)	0.113±0.050	**76.2**	0.413±0.299	76.2	1.137±0.889	81.0	2.313±1.474	61.9
Ours (400k+300k)	**0.109±0.052**	73.8	**0.349±0.248**	**81.0**	**0.916±0.576**	**88.1**	**1.895±0.888**	**88.1**
SurgT Porcine Abdomen								
Model	**Step 1**		**Step 3**		**Step 7**		**Step 9**	
	CD↓	< 0.40 **mm** ↑	**CD↓**	< 0.60 **mm** ↑	**CD↓**	< 0.90 **mm** ↑	**CD↓**	< 1.20 **mm** ↑
MGN (300k+150k)	0.468±0.682	73.8	0.515±0.473	76.2	0.803±0.647	76.2	1.142±0.890	69.0
MGN (400k+300k)	0.529±0.649	59.5	**0.515±0.481**	78.6	0.744±0.539	78.6	0.989±0.797	71.4
Ours (Scratch, 150k)	0.480±0.723	74.4	0.757±1.211	58.1	2.190±1.891	11.6	4.906±5.014	4.7
Ours (Scratch, 300k)	0.465±0.717	76.7	0.730±1.032	62.8	3.147±4.650	16.3	9.860±15.277	7.0
Ours (300k+150k)	**0.433±0.648**	78.6	0.618±1.194	**81.0**	**0.639±0.927**	**90.5**	**0.965±1.040**	**83.3**
Ours (400k+300k)	0.438±0.689	**81.0**	0.638±1.290	76.2	0.699±1.028	85.7	0.971±1.121	76.2

TABLE VI
ABLATION OVER THREE TEST TRAJECTORIES. ✓ DENOTES COMPONENT INCLUSION. CD IN MM

Model variant	Acc.	Flow	Synthetic GI (CD)			Hamlyn Silicone Phantom (CD)			Hamlyn Heart MIS (CD)		
			Step 3 ↓	Step 7 ↓	Step 9 ↓	Step 3 ↓	Step 7 ↓	Step 9 ↓	Step 3 ↓	Step 7 ↓	Step 9 ↓
Full model (ours)	✓	✓	**0.683**	**1.498**	**2.387**	**0.250**	**0.703**	1.123	**0.746**	**1.871**	**3.119**
w/o acceleration encoding	✗	✓	0.992	4.574	10.520	0.334	0.982	1.496	1.570	7.778	14.080
w/o optical flow encoding	✓	✗	1.726	8.525	11.430	0.268	0.726	**0.914**	1.042	2.967	3.802

corresponding extended-horizon quantitative results reported in Table IV.

2) Cross-Scene Adaptation: Table V reports cross-scene adaptation results across clinically distinct settings. In the Hamlyn Heart scenario, transferring between cardiac sequences with differing anatomy and motion, fine-tuning consistently improves performance across prediction steps, with gains preserved at extended horizons. For example, the 400k+300 k model achieves 0.916 mm at Step 7, corresponding to a 28.4% CD reduction over MeshGraphNet, and maintains lower error at Step 9. In the SurgT Porcine Abdomen case, where procedures span different abdominal regions, the 300k+150 k model yields stable improvements from short to extended horizons, achieving 0.639 mm at Step 7 and a 20.4% CD reduction compared with MeshGraphNet. In contrast, MeshGraphNet degrades over time under scene transfer, with reduced accuracy at longer horizons. Fine-tuning for 150 k steps completes in approximately 25 minutes on a single RTX 2080 Ti.

C. Ablation Study

Table VI reports ablation results on acceleration and optical-flow cues. The full model achieves the lowest CD across most datasets and steps, confirming their importance for stable multi-step prediction. Removing acceleration causes the largest degradation, e.g., CD reaches 14.08 mm on Hamlyn Heart at Step 9. A minor exception occurs on the Hamlyn Silicone Phantom, where removing flow slightly improves performance due to unreliable estimation in reflective, texture-poor regions.

TABLE VII
INFERENCE SPEED ACROSS GRID SIZES (FRAMES PER SECOND)

Grid	Mean	Max	Min	Std.	Latency (ms)
10×10	61.46	63.19	60.44	0.52	16.27
20×20	56.17	56.94	53.52	0.92	17.80
30×30	51.51	52.43	49.89	0.82	19.41

D. Runtime Performance

We evaluate inference speed on the GI dataset using 20 randomly sampled sequences (20 frames each) across three mesh grid sizes. Experiments are conducted on an NVIDIA RTX 2080 Ti GPU with an AMD Ryzen 9 7950X CPU. Table VII reports the FPS statistics. Across all grid sizes, throughput remains above 49 FPS with low variance, corresponding to mean latencies of 16.3 ms (10×10), 17.8 ms (20×20), and 19.4 ms (30×30). These results demonstrate stable real-time performance as mesh resolution increases.

E. Limitations

A key limitation of the proposed method is the assumption that optical flow remains reliable across the scene. In practice, flow estimation often degrades near organ boundaries, particularly in reflective, texture-poor, or occluded regions, leading to prediction errors. This is evident in boundary divergence (Fig. 3) and slight performance drops (e.g., at Step 5 on Hamlyn Silicone Phantom in Table III, and in Table VI). This limitation can be mitigated by incorporating flow confidence masks and

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by down-weighting flow cues in favour of kinematic priors when local flow quality is low. In addition, Table III shows slightly higher Step 1 error on the Synthetic GI dataset compared to MeshGraphNet. We attribute this to the auxiliary signals used in our method, which may introduce sensitivity to noise at short horizons under idealised synthetic conditions, while maintaining stability at extended horizons. green Finally, while effective at generalising across anatomically consistent subjects, our framework remains limited by significant domain gaps. If the target scene differs substantially in biomechanical properties or viewing angles from the source, the benefits of transferred priors diminish, requiring extended fine-tuning or even performing no better than training from scratch.

V. Conclusion

This letter presents a physics-informed, vision-guided, self-supervised framework greenfor capturing complex non-linear soft-tissue dynamics and making accurate deformation predictions. By integrating motion-aware encodings with physical priors, the model enables multi-step 3D forecasting from camera-only input, achieving real-time and generalisable performance across clinical scenes. Ablation studies, cross-scene evaluation, and runtime analysis validate the contribution of acceleration and flow cues and demonstrate robustness under varying surgical conditions. Our experimental results demonstrate sub-millimetre accuracy and robust long-horizon stability. This precision matches the scale of anatomical motion, ensuring the geometric consistency essential for perception-driven surgical navigation. Future work will address boundary flow reliability and extreme view adaptation.

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